

C++14 (g++ 10.3.0)



✓ Auto-complete ready!

```
1 > #include<bits/stdc++.h> ...  
3   int Consistent (int N) {  
4       // Write your code here  
5  
6       return 0;  
7 > }
```

C++14 (g++ 10.3.0)



✓ Auto-complete ready!

```
1 > #include<bits/stdc++.h> ...  
3   vector<int> solution (int N, int M, vector<int> requests) {  
4       vector<int> ans(N,0);  
5       return ans;  
6 > } ...
```

One block

You are working in the resource distribution team of your company. A *One block* is a block of data having exactly one resource which has value 1. You are given an array *Arr* containing *N* resource values.

What is the number of ways to divide the array into continuous blocks such that each block is *One block*.

NOTE - If we cannot form a block with *One block*, then output **0**.

Function description

Complete the function `OneBlock()`. This function takes the following 2 parameters and returns the required answer:

- *N*: Represents the number of resources
- *Arr*: Represents the value of resources

Input format for custom testing

Note: Use this input format if you are testing against custom input or writing code in a language where we don't provide boilerplate code

- The first line contains an integer *N* denoting the number of resources.
- The second line contains *N space-separated* integers denoting the value of resources.

Output format

Print the number of ways to divide the array into continuous blocks such that each block is *One block*.

Constraints

$$1 \leq N \leq 100$$

$$0 < Arr_i < 1$$

Consistent data

A sequence containing only 0s and 1s is called a consistent sequence if no adjacent values in the sequence are the same. For example (010101....) is a consistent sequence while (011) is not. You are allotted a simple task.

Given an integer N that represents the size of a consistent sequence starting with '0'.

Find out the number of consistent subsequences of the given sequence mod $10^9 + 7$.

Note: A subsequence is a sequence that can be derived from the given sequence by deleting zero or more elements without changing the order of the remaining elements.

Function description

Complete the function `Consistent()`. This function takes the following parameter and returns the required answer:

- N : Represents the size of the sequence

Input format for custom testing

Note: Use this input format if you are testing against custom input or writing code in a language where we don't provide boilerplate code

- The first line contains an integer N representing the sequence's size.

Output format

Print the number of consistent subsequences mod $10^9 + 7$.

Constraints

$$1 \leq N \leq 10^6$$

Sample Input



Sample output



A company has N servers, labeled from 1 to N . In the beginning, all the servers have no workload.

M requests are sent to the company's API. The company will distribute these requests evenly among its servers to keep the workload balanced. The workload is balanced when the busiest server has the least possible load.

Which server will handle the request for each API request made?

Notes

- Each request adds to the overall workload of the servers.
- If two servers have the same minimum workload, the server that is labeled with the smaller number will handle the request.

Function description

Complete the `solve()` function provided in the editor. The function takes the following 3 parameters and returns the solution:

- N : Represents the number of servers
- M : Represents the number of requests
- `requests`: Represents the load of each request

Input format for custom testing

Note: Use this input format if you are testing against custom input or writing code in a language where we don't provide boilerplate code

- The first line contains N denoting the number of servers.
- The second line contains M denoting the number of requests.
- The third line contains an array `requests` denoting the load of each request.

Output format

Print an array, denoting the server that serves each request.

Constraints

$$1 \leq N \leq 10^5$$

$$1 \leq M \leq 10^5$$

$$1 \leq \text{request}[i] \leq 10^9$$

In Machine Learning, the R-squared method is used to measure the goodness-of-fit feature of models. A feature is added to the linear regression model to train the same model again. Which of the following statements about the R-squared value is correct?

- ☐ If the R-Squared value increases, then the variable of the model remains significant.
- ☐ If the R-Squared value decreases, then the variable of the model remains insignificant.
- ☐ The R-Squared value cannot determine the variance of a model.
- ☐ None of these

[Reset Answer](#)

Search





End

Which of the following statements about GloVe is true?

- ☐ GloVe represents words as dense vectors by modeling the relationships between words using only local contexts.
- ☐ The loss function in GloVe is designed to minimize the squared difference between predicted and actual co-occurrence probabilities.
- ☐ GloVe uses a fully connected neural network to learn word vectors from a co-occurrence matrix.
- ☐ GloVe and Word2Vec use identical algorithms for generating word embeddings.

Reset Answer



Search



In Machine Learning, you are working on a project using a neural network. You are using the rectified linear (ReLU) activation function. Which of the following statements about the ReLU activation function is correct:

1. It is a piecewise linear function
2. If it is positive, it outputs the input directly. Otherwise, it outputs zero.
3. It is capable of providing a true zero value
4. It acts as a linear activation function.

☐ 1 and 2

☐ 2 and 3



Q Search



15
16
17
18
19
20
21
22
23

John is working on a binary classification problem and needs help understanding what it means if his AUC score is nearly zero while working on an application. Can you help him find the appropriate information?

☐ Model is reciprocating the result.

☐ Model is performing well.

☐ Model is overfitting.

☐ None of these





5 /
23 Completed

01:13:42



John is working on Machine Learning. While working on an application he wants to know the cost function for using logistic regression modeling. Help him to find which of the following can be used in the above-mentioned context.

Options:

1. $\frac{-1}{m} \left[\sum_{i=1}^m y^{(i)} \log(h\theta(x^{(i)})) + (y^{(i)}) \log(1 - h\theta(x^{(i)})) \right]$

2. $\frac{-1}{m} \left[\sum_{i=1}^m y^{(i)} \log(h\theta(x^{(i)})) + (1 - y^{(i)}) \log(1 - h\theta(x^{(i)})) \right]$

3. $\frac{-1}{m} \left[\sum_{i=1}^m y^{(i)} \log(h\theta(x^{(i)})) + (1 - y^{(i)}) \log(h\theta(x^{(i)})) \right]$

4. $\frac{-1}{m} \left[\sum_{i=1}^m \log(h\theta(x^{(i)})) + (1 - y^{(i)}) \log(1 - h\theta(x^{(i)})) \right]$

☐ 1



Q Search



Question 15

Max. score: 2.00

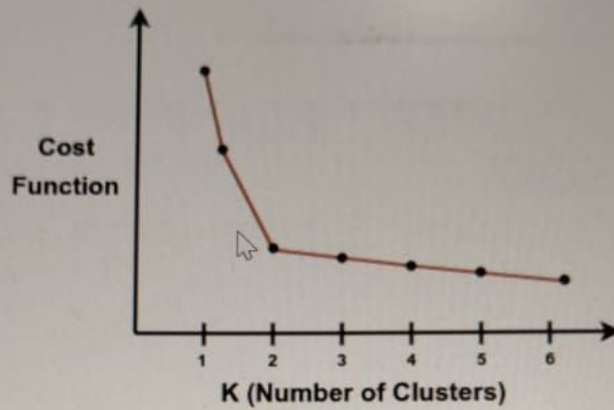
Artificial Neural Network (ANN) is also comprehended as Universal Function Approximators. Which of the following is one of the important reasons behind this?

☐ Weights☐ Activation function☐ Sum of the inputs used during training☐ Loss[Reset Answer](#)

Q Search

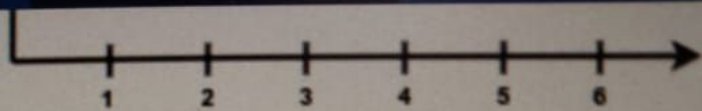


In Machine Learning, you are working on an unsupervised algorithm called the **k-means** algorithm. One of the important steps in this algorithm is to determine the optimal number of clusters into which the data may be clustered. From the given graph find the value of **K** that represents the number of clusters:



☐ 1





K (Number of Clusters)

☐ 1

☒ 2



☐ 3

☐ 4

 Reset Answer



Q Search



Question 13

Max. score: 2.00

Anita built a machine learning model that initially worked smoothly. However, the model suddenly suffered from high variance and faced low bias. Which supervised learning algorithm can she use in this scenario to tackle high variance?

☐ Classification☐ Linear Regression☐ Naïve Bayes☐ Random Forest

Q Search





1. This is the layer where a majority of the computations occur. Here, the computer distinguishes between the white pixels as 1 and black pixels as -1.
2. This is the layer where a majority of the computations occur. Here, the computer distinguishes between the white pixels as 0 and black pixels as 1.
3. When you input an image through this layer, a model extracts the unique feature of an image such as colors, shapes, and edges that defines the image. Once you have the features of the image, you perform the filtering process.
4. This layer produces multiple feature maps over different iterations.

☐ 1, 3, and 4☐ 2, 3, and 4☐ 1 and 3

Q Search



☐ 1, 2, and 3

☐ 2, 3, and 4

☐ 1, 3, and 4

☐ All of these

[Reset Answer](#)



Q Search



assessmenthacker.com/challenges/...
All Bookmarks

h 17 / 23 Completed 00:49:26 al21@iitk.ac.in ? End

Question 9 Max. score: 4.00

John is given the eight actual values of the target variable in the train file of a Machine Learning model. The values are as follows:

[0, 0, 0, 0, 1, 0, 1, 1]

According to the given values, what is the entropy of the target variable?

☐ $-(5/8 \log(5/8) + 3/8 \log(3/8))$

☐ $(5/8 \log(5/8) - 3/8 \log(3/8))$

☐ $(5/8 \log(3/8) - 3/8 \log(5/8))$

☐ $-(5/8 \log(3/8) + 3/8 \log(5/8))$

Search



A company wants to track the number of calls they receive over a day to decide how many people they want to hire. They make the following assumptions.

You are given a task to determine the probability of receiving exactly k number of calls during a specific day. Which of the following distributions can you use to complete this task:

Assumptions

1. Each call occurs independently from every other call.
2. A given call has the same probability of occurring at any given time of the day.
3. Calls occur at a rate of λ calls per day.

☐ Binomial

☐ Poisson



Q Search

